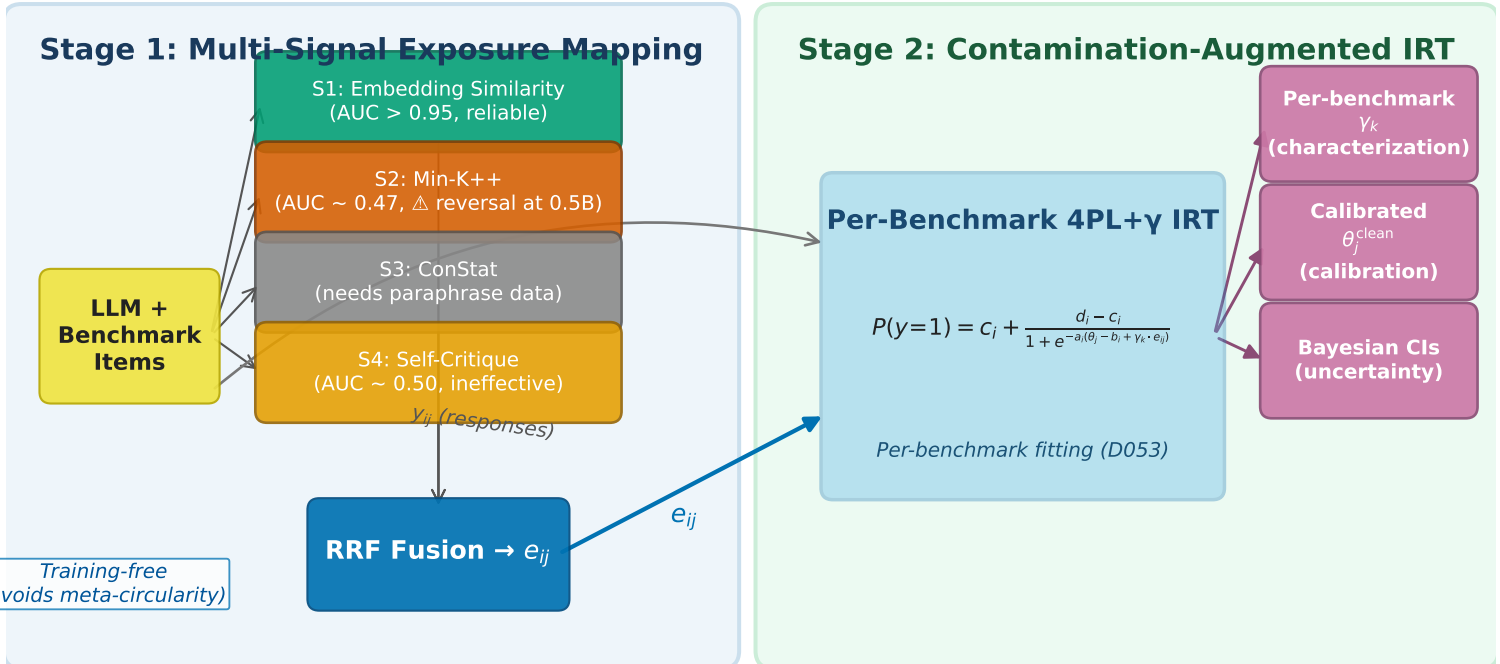
Stage 1: Multi-Signal Exposure Mapping
S1: Embedding Similarity
(AUC > 0.95, reliable)
S2: Min-K++
(AUC ~ 0.47, ⚠ reversal at 0.5B)
S3: ConStat
(needs paraphrase data)
S4: Self-Critique
(AUC ~ 0.50, ineffective)
LLM + Benchmark Items
y_{ij} (responses)
RRF Fusion → e_{ij}
e_{ij}
Training-free
(avoids meta-circularity)
Stage 2: Contamination-Augmented IRT
Per-Benchmark 4PL+γ IRT
$P(y=1) = c_i + \frac{d_i - c_i}{1 + e^{-a_i(\theta_j - b_i + \gamma_k \cdot e_{ij})}}$
Per-benchmark fitting (D053)
Per-benchmark γ_k (characterization)
Calibrated θ_j^{clean} (calibration)
Bayesian CIs (uncertainty)